



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

Accredited by NAAC & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering

Academic Year 2023 – 2024

Minutes of Meeting – 6th Department Advisory Committee (DAC)

Date of Meeting : 26.09.2023 (Tuesday)
Time of Meeting : 02:00 PM to 4:00 PM
Reference : RIT/Mech./DAC/06
Venue : Mechanical Department Seminar Hall
Frequency of meeting: Once in a year

Major functions of Department Advisory Committee meetings are

- Review of Academic plans preparation by the department, Student performance, Faculty contributions, stakeholders feedback and department progress.
- Observe the process Curricular Gap/Content beyond the syllabus to meet PEOs, PSOs and POs.
- Appraisal of Student support system Co-curricular / Extra-Curricular activities
- Monitor the status of continuous improvement.

The meeting was convened with the following agenda:

- Review of action taken & Report of the previous meeting
- Review of stake holders / Summary of feedback collected
- Program curriculum and Teaching & Learning process
- Incorporating Anderson-Bloom-Vincenti (ABC) Taxonomy table
- Course outcomes and program outcomes
- Student performance and faculty contribution
- Continuous process improvement
- Any others (Transformation of affiliation status to Autonomous status/ Implementation of CDIO Framework/ Research centre recognition/ NBA Accreditation-Compliance status for NBA 2nd cycle/ Adapting functional framework for accreditation and ranking)

The members were present in the meeting are given below.

Members present:

- 1) Mr.T.Thirumalai kumar, Manager HR, Flender India Pvt. Ltd., Chennai. (Attended thro' Online)
- 2) Dr.N.Durairaj, Additional General Manager, Boiler Shops Unit II, BHEL, Tiruchi. (R&D Member) (Attended thro' Online)
- 3) Dr.R.Jeyapaul, Prof. and Head, Department of Production Engineering, National Institute of Technology, Tiruchirappalli (Academic

2. Dr.K.Karthikeyan, ASCP/EEE
3. Dr.M.Gomathy Nayagam, ASCP/CSE
4. Dr.O.Senthilkumar, ASCP/Chemistry
Dr.T.Manimaran, ASCP/Maths
5. Mr.T.Chockalingam, AP(SG)/Civil
6. Dr.P.Sureshkumar, ASCP/Mech.
7. Dr.V.Sivakumar, ASCP/Mech.
8. Dr.J.Jerold John Britto, AP(SG)/Mech.
9. Mr.R.Arun Kumar, AP(SG)/Mech., RIT
10. Mr.S.P.Cheran (Reg. No. 953620114006), IV

Member)	Year Mech.
4) Dr.P.Maran, Professor, Department of Mechanical Engineering, TCE, Madurai, Academic Member.	11. Mr.M.Jeeva (Reg. No. 953622114012), II Year Mech.
5) Mr.A.Arun Shenbaga Raj (Batch 2016-2020) Founder & CEO, TN76 Food Delivery Service, Tenkasi. (Attended thro' Online)	12. Mr.P.Sivaharani Priya (Reg. No. 953622114024), II Year Mech.
6) Mr.A.Thulasiram, DGM-Factory Head, Gowri House Metal Works LLP, Rajapalayam. Industry Member.	Faculty Members – Dept. of Mech. Engg., RIT
Mr.A.R.Jaisaravana, Workshop Superintendent, Gowri House Metal Works LLP, Rajapalayam. Special Invite.	1) Dr.M.Lakshmanan, AP(Sr.Gr.)
7) Mr.A.Krishna Kumar (Batch 2016-2020) Alumni, - Special Invitee.	2) Mr.M.Ashok Kumar, AP(Sr.Gr.)
Internal Members from RIT	3) Mr.C.Gururaj, AP(Sr.Gr.)
1. Dr.S.Rajakarunakaran, Prof & Head/Mech.	4) Mr.M.Sivagaminathan alias Balaji, AP(Sr.Gr.)
	5) Mr.G.Prabu ram, AP(Sr.Gr.)
	6) Mr.S.Maharajan, AP(Sr.Gr.)
	7) Mr.S.Valai Ganesh, AP (Sr.Gr.)
	8) Mr.L.Karthikeyan, AP (Sr.Gr.)
	9) Mr.R.Venkatesh, AP
	10) Mr.M.Ramar, AP
	11) Mr.S.Kathiravan, AP

Leave / On Duty

S.No.	Name of the member	Designation	Position	CL/OD/ Remarks
1.	Mr.V.K. Vikram	(Batch 2015-2019) Systems Engineer, Infosys Limited, Technopark Campus, Tirvandrum, Kerala	Alumni Member	Official work
2.	Mr.S.Shanmugam	(Batch 2015-2019) Senior Executive, Rane TRW Steering Systems Pvt Ltd. Alwarpet, Chennai.	Alumni Member	Official work
3.	Mr.D.Sree Balasubramanian,	F/o S.Pradeep durai (IV Year B.E. Mech.) Office Superintendent, BSNL, Nagarcoil.	Student Parent	Official work

Mr.M.Ramar, AP/Mech. welcomed all the DAC members.

Dr.S.Rajakarunakaran, Prof. and Head / Mech. presented the 6th DAC meeting agenda as follows:

1. Review of action taken & Report of the previous meeting

1.1 NBA Compliance report – Second Cycle

Briefed the action taken for the NBA compliance report as follows:

Suggested weakness	Action Taken
Experts from academic & industry are being invited for guest lectures.	Conducting guest lecture through professional societies, MoU signed companies, and Extra Mural Lecture series.

Faculty / Students / Non-Teaching staff are encouraged to attend FDP, STTP / Intern / Training Programs	Faculty members, students, and non-teaching staff member are developing their cognitive, inter personal and psychomotor skills by attending various training programmes.
Students are encouraged to participate in competitions	Students are participating in Smart India Hackathon, India Skills competition, Tamil Nadu Skill Development, YUKTI innovation challenge, and internal Hackathons.
Student enrollment is dropping each year	Disseminated the credentials of the department and achieved 96.67 % admission in the current AY 2023-24
Less number of students go for higher studies & entrepreneurship	Career Guidance Cell, Alumni association and Entrepreneurship Cell is thriving and progressing well
More industry-oriented projects should be given to students	Students are motivated to do project in MoU signed companies.
More faculty member is encouraged to get sponsored & consultancy projects	Conducting workshop/seminar/orientation programme related to project proposal submission and receiving research mentorship from adjunct and honorary professors.

1.2 5th DAC Meeting Agenda 10.09.2022)

Agenda	Points discussed
1	Review of Previous meeting (4 th DAC, 4 th & 5 th PAQIC)
2	Teaching learning process
3	Impact analysis of industry institute interaction
4	Impact analysis of industrial training/ Internship and feedback analysis
5	Initiatives taken in the target set for the attainment of COs/POs/PSOs and addition of Co-curricular and Extracurricular in PO
6	Student Participation (co-curricular and extra-curricular activities)
7	Research and Development (Academic Research, Sponsored Research, Product Development, Research laboratories)
8	Academic Performance Indicators based on Performance Based Appraisal System
9	Faculty participation and contributions
10	Visiting faculty - progress, Status of continuous improvement, and Academic audit implementation and effectiveness

1.2.1 ACTION TAKEN REPORT (5th DAC Meeting on 10.09.2022)

Activities/Tasks proposed	Action Taken
- Construct a lab with state of art with benchmarking - Students should get exposure in mechanical along with electronics engineering field (exposure in Mechatronics)	- TRCL Supported Yaskawa Robotics Laboratory at Ramco Institute of Technology is home to the state-of-the-art Motoman GP12 robot, representing a pinnacle of technology in the field of robotics. - Providing training to the students in the mechatronics field and encourage them to participate in state/national/ and international level competitions.
- Improved communication skill set is desired for getting placement in MNCs. - Provide training to understand the "basic quality tools usage in industries" to both software and core company placed students.	Through department placement coordinator the following activities are conducted i) training for communication skill ii) Group discussion iii) Mock interview and iv) company specific interview. <i>It is a continuous process</i>
- Introduce new electives and create student interaction with industry person for continuous improvement. - Modern technology-based courses can be given as a value-added course.	For third year students' <i>elective courses</i> i) CME362 Energy Conservation in Industries ii) CME385 Refrigeration and Air Conditioning iii) CME387 Nontraditional Machining Process iv) CME397 Surface Engineering v) CME365 Renewable Energy Technologies and vi) CME380 Automobile Engineering <i>are offered in the emerging field</i>
- Assessment tools can be improved for Teaching and learning process - Categorize teaching strategies, metrics for measuring the strategies	- Students are grouped based on their academic performance. - Peer learning is encouraged in which bright learners help the slow learners. Performance of the students are assessed based on the group they fall.
- Give more importance to extra-curricular activities. - Student participation in co-curricular and extra-curricular activities can be measured in-terms of their batch size. Proper rubrics should be followed in the student project assessment evaluation.	- Celebrated <i>sports day</i> during 10th March 2023 - Without fixing any constraint, motivating the deserved students to engage with co-curricular and extra-curricular activities. - As per <i>AICTE examination reform policy</i>, the project rubrics are developed and it is followed for Batch 2019 – 2023 onwards
Insisted to produce 10% entrepreneurs from a batch of students and 30% projects should be real time implementations.	Regularly conducting entrepreneurship awareness programme through EDC and IIC. <i>54 Events conducted. Received 1 project grant from IE(I), 3 from TNSCST, 3 idea grants from MSME during this period</i>

• Internship credits can be given to the students. Internship period should be at least one to two months. Permit the students to do certification courses offered by the industry.	• As per AU R2021 internship is mandatory for all the students. Through internship coordinator the internship activities are monitored and appropriate methodology is adopted for internship assessment. Mr.S.P.Cheran, IV Year Mech. is completed his <i>one and half month internship at IITM.</i>
• Inclusion of weightage for Co-curricular activity in CO attainment	• <i>Based on the PAQIC members suggestions, it has proposed to consider 10% weightage for co-curricular activities (based on the student's participation in other states, within states, or internship participation) in PO attainment.</i>
• Give awareness program on govt. opportunities and recent technology awareness for pursuing their career.	• Student Alumni Interaction on 17.06.2023 "<i>Career Awareness meet Smart Tips to Crack Government Exams</i>" - Mr.Mohamed Musthafa S (Alumni 2013-2017 Batch) Lecturer, Dept. of Mechanical Engg, Government Polytechnic College, Chekkanurani, Madurai • Career Guidance on-17.07.2023 "<i>Defense and Core Industry</i>"- CAPTAIN M. B. VYAS NARAYANA, (2013-2017 batch) Indian Army, Mr. K. GOPINATH, (2013-2017 batch) Mercedes Benz research and development India, Bangalore • My Story in RIT, Mr.N.Siranjeevi, (Alumni 2017-2021 Batch), Assistant Commandant (Technical), Indian Coast Guard, India. https://www.ritjpm.ac.in/departments/mechanical-engg/mechalumni-interaction.php
• Final year project work can be assigned during II-year course of study itself. • Motivate the students to do real time project in industry The outcome from the project work should be improved and Hackathon can be conducted for solving problems	• Students are encouraged to participate in Smart India Hackathon, India Skills competition, YUKTI innovation challenge, and internal Hackathons. The students are encouraged to take the presented ideas as their academic projects.
• MoU with multinational companies may directly support to improve the perception about the institute and indirectly helps to achieve NIRF ranking.	• The effectiveness of MoU is measured through IIC and proposed to sign MoU with multinational companies.
• Improve the student counseling system	• <i>For each faculty mentor, it has been proposed to assign mentees from I, II, and IV year students</i>

1.3 6th PAQIC Meeting Agenda (07.03.2023)

Agenda	Points discussed
1	Review of Previous meeting (6 th PAQIC)
2	New initiatives taken for quality improvement in Teaching and Learning – Outcome perspective
3	Impact analysis and effectiveness for: Quality of internal semester question paper, Assignment, Evaluation, industrial training and feedback format, mentoring system
4	Initiatives for adding Co-curricular and Extra-curricular activities towards CO attainment
5	Proposed methodologies to set target attainment for CO i) Design Domain and ii) Manufacturing Domain R 2021
6	Student's performance in Co-curricular and Extracurricular activities
7	Progress in R&D and institute industry interaction cell progress
8	Conclusion remark of the academic audit
9	Department Outreach Activities

1.3.1 ACTION TAKEN REPORT (6th PAQIC Meeting on 07.03.2023)

Activities/Tasks proposed	Action Taken
Derive a minimum of two projects as an outcome from the course "Advances in Manufacturing Engineering."	The outcomes are generated in terms of projects as well as internship. Three students are recommended to do internship at IITM, based on the guidance from the Adjunct Professor, Dr.L.Vijayaragavan, (Retired IITM). Mr.S.P.Cheran, IV Year Mech. is shortlisted and he completed his one and half month internship, also he is recommended to do his final year project at IITM.
The subjects' pre-requisite tests should be divided into two categories: i) relevant/related subject content already studied, and ii) relevant/related subject content not exposed. Elective courses may not be subjected to pre-requisite testing.	All the subject handling faculty members are conducting prerequisite test.
Collect feedback during the conduct of any kind of active learning in the teaching and learning process.	Feedback were collected and analysed https://www.ritrjpm.ac.in/images/mechanical/2022-2023/ME8691_%20CAD_CAM.pdf
Appreciate the students and faculty members for their achievements.	Appreciation for the students and faculty members are done through the department social media.
Allocation of weightage for the course Performance in the previous semester may be considered at 40% instead of 50%. The remaining 10% weightage may be given to the prerequisite test.	For R2017 course Previous Semester Grade Point Average (GPA) of respective subject For R2021 course Prerequisite Course Semester Result (GPA) of the same batch students (30%)

	• Number of Times Subject handled (5%) • Faculty Experience (5%) • Previous two semesters result (GPA) – Respective Subject (40%) Average Pass Percentage of previous two semesters results (Class Result in %) (20%)
Identify the potential students from the first year itself and motivate them to secure a university rank.	Based on the 1st semester university examination performance the following students are identified and motivated to secure a university Rank. (Batch: 2022-2026) • JEEVA M (953622114012) (8.36) • SANJITHKUMAR P (953622114030) (8.05) • P SIVAHARANI PRIYA (953622114024) (8.00)

2. Review of stake holders / Summary of feedback collected

Listed out the salient feedback collected from various stakeholders during Department Review meeting, Class Committee and Course Committee Meeting, Parent Teacher Meeting, Governing Council Meeting, and Interaction with Industry experts.

<i>Faculty and Staff</i>	<i>Students/Alumni</i>	<i>Parents</i>	<i>Management/ Governing Council</i>	<i>Industry</i>
• Career growth, • Research facilities & incentives • Technology based academic & working ambience • Academic flexibility • independence with accountability	• Support for co-curricular & Extracurricular activities • State of the art infrastructure • Experiential learning & Opportunity for talent exposure • International learning at affordable cost • Quality Placement, career guidance and entrepreneurial opportunities	• Good placements • Quality teaching-learning • Disciplined students • Institution Brand/ Reputation	• Autonomous Status • Good E-Governance for all process • Good NIRF ranking with in top 100 in India • Competent Faculty in emerging areas • Internal Revenue for sustainability through R&D Projects/ • Consultancy and other IP Assets • Offering Industry oriented Courses • Bench marking through Accreditation of	• Industry ready professionals • Strong fundamentals • Strong Industry-Institution interaction • Collaborative research • consultancy • NAAC Accreditation and NIRF ranking of the institute • NBA Accreditation with good score

			programs and institution	
			• Creation of Centres of excellence	

3. Program curriculum and Teaching & Learning process

3.1 Detailed the role of domain coordinator in the curriculum gap identification & implementation in the following subjects.

- Thermal Domain:
 - CME385 Refrigeration and Air Conditioning
 - CME 362 Energy Conservation in Industries
- Design Domain:
 - CME340 CAD/CAM
- Manufacturing Domain:
 - ME3393 Manufacturing Processes
 - CME387 Non-Traditional Machining Processes
 - MG8091 Entrepreneurship Development

3.2 Steps taken towards Deepening - Honorary Courses & Knowledge Acquisition

Nomination of verticals in-charges

- Modern Mobility Systems: Mr.M.AshokKumar AP(SG)/Mech.&Mr.L.Karthikeyan AP(SG)/Mech.
- Product and Process Development: Dr.J.Jerold John Britto, AP(SG)/Mech.
- Robotics and Automation – Mr.S.Valai Ganesh, AP(SG)/Mech.

3.3 Listed out the internship details (AY 2022-2023)

Sl. No.	YEAR	No of students	Completed two weeks mandato internship	Yet to complete the internship
1	IV	84	75	9
2	III	63	63	0
3	II	55	-	-

3.4 Listed out the domain wise innovative practices carried out

3.4.1 Design and Allied Courses

Sl. No.	Subject Code & Name	Name of the Faculty Members	Topic	Name of the Innovative Practice
1.	ME8099 Robotics	Mr. S. Valai Ganesh, AP (SG)/Mechanical	Simple Circuit with Tinker CAD	Group Challenge
2.	ME8099 Robotics	Mr. S. Valai Ganesh, AP (SG)/Mechanical	Homogenous Transformation Matrix Formulation	Software Simulation
3.	ME8691 Computer Aided Design and Manufacturing	Dr.J.Jerold John Britto, AP (SG)/Mechanical	Simulate the Data exchange formats of IGES, STEP, STL., using standard CAD Models	Theory to Practice

3.4.2 Thermal and Allied Courses

Sl. No.	Subject Code & Name	Name of the Faculty Members	Topic	Name of the Innovative Practice
1.	ME8792 Power Plant Engineering	Dr.V.Sivakumar, ACSP/ Mechanical Mr.M.Ashok Kumar, AP (SG)/Mechanical	Diesel, Gas Turbine and Combined Cycle Power Plants	Theory to Practice
2.	ME3391 Engineering Thermodynamics	Mr.R.Arun Kumar, AP/Mech & Mr. M. Sivagaminathan alias Balaji, AP(SG)/Mech	Properties of Pure Substances / Calculation of Steam properties	Collaborative Learning
3.	ME8693 Heat and Mass Transfer	Mr.M.Ashok Kumar, AP (SG)/Mechanical	Unit I to Unit V – All Tutorial Classes	Peer Learning

3.4.3 Manufacturing and Allied Courses

Sl. No.	Subject Code & Name	Name of the Faculty Members	Topic	Name of the Innovative Practice
1.	ME3493 Manufacturing Technology II	Mr.S.Maharajan, AP(Sr.Gr)/Mech	Capstan and Turret Lathe, Tumbler Gear Mechanism, Apron Mechanism and Half Nut Mechanisms	Theory to Practical
2.	GE8077 Total Quality management	Dr.S.Rajakarunakaran, Prof. and Head/Mech Mr.G.Prabu ram, AP(Sr.Gr)/Mech	1. Customer Satisfaction 2. Team and Team Work	Coolmath games- Coffee shop, Guide the Blind, Hula Hoop & Water Relay Race

4. Incorporating Anderson-Bloom-Vincenti (ABC) Taxonomy table

Presented the Anderson-Bloom-Vincenti Taxonomy table and requested the DAC member to give permission to incorporate this practice from the forthcoming semester onwards.

Anderson-Bloom-Vincenti Taxonomy Table

Cognitive Processes	Knowledge Categories							
	Factual	Conceptual	Procedural	Metacognitive	Fundamental Design Principles	Criteria & Specifications	Practical Constraints	Design Instruments/Tools
Remember								
Understand								
Apply								
Analyze								
Evaluate								
Create								

5. Course outcomes and program outcomes

Detailed the CO-PO Course Articulation Matrix Mapping with Course Name: ME3591 Design of Machine Elements

Type	POs	Action Verb(s) in POs	Bloom's level(s) for POs	Bloom's level(s) for COs
Technical	PO 1	Apply	L3	• Bloom's L1 to L4 for Theory Courses. • Bloom's L1 to L5 for Laboratory Courses. • Bloom's L1 to L6 for Mini Project and Main Project.
	PO 2	Identify	L2	
		Formulate	L6	
		Review	L2	
	PO 3	Design	L6	
		Develop	L3, L6	
	PO 4	Analyse	L4	
		Interpret	L2, L3	
		Design	L6	
	PO 5	Create	L6	
		Select	L1, L2, L6	
		Apply	L3	
Non-Technical	PO 6	THUMB RULE		
	PO 7	• If Bloom's L1 Action Verbs of a CO: Correlates with any of PO6 to PO12, then assign 1.		
	PO 8	• If Bloom's L2 to L3 Action Verbs of a CO: Correlates with any of PO6 to PO12, then assign 2.		
	PO 9	• If Bloom's L4 to L6 Action Verbs of a CO: Correlates with any of PO6 to PO12, then assign 3.		
	PO 10			
	PO 11			
	PO 12			

CO.No.	Course Outcomes	Po's	Cognitive Level	Knowledge Category	Classes
CO 1	Explain the design machine members subjected to static and variable loads.	1,2,3,8,9 & 12	Un	F & C	12
CO 2	Apply the concepts design to shafts, key and couplings.	1,2,3,8,9 & 12	Ap	F & C	12
CO 3	Apply the concepts of design to bolted, Knuckle, Cotter, riveted and welded joints.	1,2,3,8,9 & 12	Ap	F & C	12
CO 4	Apply the concept of design helical, leaf springs, flywheels, connecting rods and crank shafts.	1,2,3,8,9 & 12	Ap	F & C	12
CO 5	Apply the concepts of design and select sliding and rolling contact bearings, seals and gaskets.	1,2,3,8,9 & 12	Ap	F & C	12
CO 6	Evaluate the various stresses available in shaft elements using MATLAB and Analyzing the Flange coupling using ANSYS	1,2,3,5,9 & 12	An	F, C, P & M	6

Un-Understand, Ap-Apply & An-Analysis

General Categories: Factual (F), Conceptual (C), Procedural (P), and Metacognitive (M)

Course Outcomes (COs)	Program Outcomes (POs)												Program Specific Outcomes (PSOs)			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4
CO 1	√	√	√	-	-	-	-	√	√	-	-	√	√	-	-	√
CO 2	√	√	√	-	-	-	-	√	√	-	-	√	√	-	-	√
CO 3	√	√	√	-	-	-	-	√	√	-	-	√	√	-	-	√
CO 4	√	√	√	-	-	-	-	√	√	-	-	√	√	-	-	√
CO 5	√	√	√	-	-	-	-	√	√	-	-	√	√	-	-	√
CO 6	√	√	√	-	√	-	-	-	√	-	-	√	√	-	-	√

Mapping Ratio

PO	CO	Mapping Ratio (%)
(1,2,3,8*,9&12)	(1 to 6)	$66^*/66=100$ Mapping Strength = 3 6/66=9 Mapping Strength = 1
5	6	

Rubrics

PO	CO	Mapping Ratio (%)
(1,2,3,8*,9&12)	(1 to 6)	$66^*/66=100$ Mapping Strength = 3 6/66=9 Mapping Strength = 1
5	6	

Percentage of Key Competencies for CO – PO Mapping

Course Outcomes (COs)	Program Outcomes (POs) (%)												Program Specific Outcomes (PSOs) (%)			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4
CO 1	67	40	50	0	0	0	0	100	50	0	0	50	100	0	0	67
CO 2	67	40	50	0	0	0	0	100	50	0	0	50	100	0	0	67
CO 3	67	40	50	0	0	0	0	100	50	0	0	50	100	0	0	67
CO 4	67	40	50	0	0	0	0	100	50	0	0	50	100	0	0	67
CO 5	67	40	50	0	0	0	0	100	50	0	0	50	100	0	0	67
CO 6	67	40	50	0	100	0	0	0	50	0	0	50	100	0	0	100

Course Outcomes (COs)	Program Outcomes (POs)												Program Specific Outcomes (PSOs)			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4
CO 1	3	1	2	-	-	-	-	3	2	-	-	2	3	-	-	3
CO 2	3	1	2	-	-	-	-	3	2	-	-	2	3	-	-	3
CO 3	3	1	2	-	-	-	-	3	2	-	-	2	3	-	-	3
CO 4	3	1	2	-	-	-	-	3	2	-	-	2	3	-	-	3
CO 5	3	1	2	-	-	-	-	3	2	-	-	2	3	-	-	3
CO 6	3	1	2	-	3	-	-	-	2	-	-	2	3	-	-	3
TOTAL	18	6	12	-	3	-	-	15	12	-	-	12	18	-	-	18
AVERAGE	3.0	1.0	2.0	-	3	-	-	3	2	-	-	2	3	-	-	3

Course Articulation Matrix Rubrics

$0 - 0 \leq C \leq 5\%$	No Correlation
$1 - 5 \leq C \leq 40\%$	Low / Slight
$2 - 40 \leq C \leq 60\%$	Moderate
$3 - 60 \leq C \leq 100\%$	Substantial / High

Allocation of Weightage for R21 Subjects

S.No.	Parameters	Allocated Weightage (%) For Theory Courses	Allocated Weightage (%) For Practical Courses	Criteria Followed
1.	Performance in Prerequisite Course, If any*	30%	-	Considering the Grade Point Average (GPA) of Prerequisite Course Semester Result of the same batch students (Average GPA of prerequisite courses will be converted into 3 scale)
2.	Number of Times Subject handled	5%	5%	Less than 2 times handled – 50% weightage (1.5) 2 to 4 times handled – 75% weightage (2.25) Above 4 times handled – 100% weightage (3)
3.	Faculty Experience	5%	5%	Less than 5 years experience - 50% weightage (1.5) Equal to and greater than 5 and less than 8 Years - 75% weightage (2.25) Equal to and greater than 8 Years - 100% weightage (3)
4.	Course Performance in Last two semester*	40%	50%	Average of Previous two semesters result (GPA) and convert into 3 point scale. (If there is a new course, the minimum target value is set to be – 1.95 for Theory and 2.40 for Practical Courses (for Theory - 21 regulation - First Class % – 65% - 3 Scale Value – 1.95) (for Practical - 80% - 3 Scale Value – 2.40))

5.	Student Quality	20%	40%	Average Pass Percentage of previous two semesters results (Convert into 3 Point scale)
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6. Student performance and faculty contribution

6.1 Listed out the student participation in an international level co-curricular and extra-curricular activities

- Mr.Gowtham Raj K (Batch 2014-2018) – Pursuing M.Sc Engineering Management – Blended in Arden University, Germany
- Mr.Sibish Rahul B (Batch 2014-2018) Completed M.Sc. Supply Chain Management and Purchasing in SKEME Business School, Francis
- Mr.Sanju S (Batch 2014-2018) completed master's in Advanced. Optional Technologies in Friedrich Alexander University of Erlangen-Nuremberg, Germany
- Mr.Deepak Krishna S (Batch 2016-2020) pursuing higher studies in UK
- Mr.Vijay Jaganath P R (Batch 2019-2023) planning to pursue his higher study in Germany.
- Mr. Hari Haran K (Batch 2016-2020) – Represented Team India in Continental Football League Moscow 2019

6.2 Fig. 6.1 shows the number of skills, number of students participated and medals won in the India Skills competition.

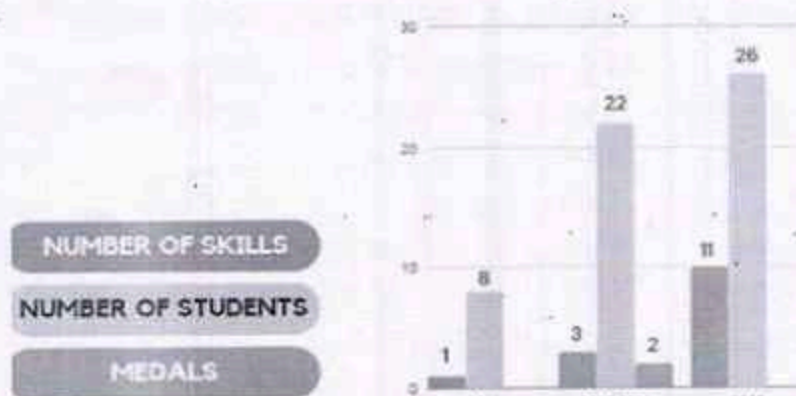


Fig. 6.1. Participation in India Skills Competition

Proposed plan to improve the international event

- Establish a mentorship program
- Integrate international events into curriculum
- Form a student international events committee
- Organize mock events

6.3 Summarized the faculty participation during this period

- No. of guest lectures delivered - 23
- Faculty presentation in National and international conferences – 7
- Faculty participation in FDP – 12
- Event organized – 19
- Ph.D. Viva voce & confirmation completed – 3
- Underwent Industrial visit – 3
- Acted as a reviewer in Journal – 7
- No. of Journal publication (SCI/Scopus) – 17
- No. of Conference publications (Scopus) - 23
- Major funded project proposal submitted – 3

- H-index – 15 (Vidwan Portal)
- Awards and patent granted – 4
- 3 Ideas granted from MSME (Rs. 31,82,800/-)
- 1 research grant received from IE(I) (Rs.32,000/-)
- 3 funded projects from TNSCST – (Rs.21,500/-)
- 2 patents filed
- 1 FER responded
- 1 copy right registered
- Received Anna University Research Centre Recognition
- Dr.J.Jerold John Britto AP(SG)/Mech. and Dr.S.Mahrajan, AP(SG)/Mech. completed their Ph.D.
- Mr.S.Kathiravan, AP/Mech. registered part time Ph.D. and doing course work.
- Mr.R.Umaselvaraj, M.E. (Outside RIT) submitted application for full time Ph.D.
- Hosted 1st International Conference on Contemporary Research Advancements in Mechanical Engineering – 24, 25 March 2023
- Dr. S.Rajakarunakaran, Prof. and Head/Mech. - ranked 369th Place in SCI Val- Elsevier Rankings and received AD Scientific Index 2023.

6.4 Effectiveness of Industry supported Laboratory and MoU signed Industries

- Mr.Maharajan S, Assistant professor (SG), has visited **Spraymet Coating Industries, Bangalore** to learn about the surface coating process on 31.07.2023.
- Mr.Arun Kumar R, Assistant professor (SG), Mr.Karthikeyan L, Assistant Professor (SG), and Mr.Prabu ram G, Assistant Professor (SG) have completed 3 – days faculty industrial Training programme in **Quantanics TechServ Pvt. Ltd., Madurai** from 20.07.2023 to 22.07.2023.
- Mr.Venkatesh R, Assistant professor, has completed 3 – days faculty industrial Training programme in **Systems and Automations, Madurai** from 17.07.2023 to 19.07.2023.
- Mr.S.Valai Ganesh, Assistant professor (SG) and Mr.C.Pandimuneeswaran have completed 3 – days faculty industrial Training programme in **YASKAWA India Pvt. Ltd., Robotics Division, Bangalore** from 24.07.2023 to 26.07.2023.

 • One student 3 Weeks Internship & Industrial Training Programme • Consultancy work of Rs. 14,550 /- completed to the industry Skillwave Technologies, Chennai	 • Faculty Industrial visit on 25.05.2023 • 6 students attended 2 weeks internship programme • FDP Industrial visit on 20.07.2023
 • One day Industrial visit - ME3493 Manufacturing Technology on 18.03.2023.	 • TVS – ATP Orientation - III Year on 06.05.2023 • 3 students completed 30 day slot – III ATP • MSME Project progress visit on 08.09.2023 • TVS-TSEV Lab visit on 08.09.2023
 • One student completed final year project from 14.02.2023 to 30.04.2023. • One student – II Year undergone Internship from 19.06.2023 to 04.07.2023. • 3 students completed final year project from 21.02.2023 to 13.05.2023	 • One student completed final year project on 25.04.2023

6.5 Industry involvement in program design and curriculum

Regulation	Subject code and Title	Detail of Industry Expert
R2021	CME380 – Automobile Engineering	Mr.M.Vignesh, Senior Engineer, Robert Bosch & S.Anbarasan, Team lead, R& D, Daimler Benz.
R2021	CME340 - CAD/CAM	M.Udaya Shankar Prabhu, Head – Technical Support, CAE Centre.
R2021	CME362 - Energy Conservation in Industries	Mr.P.Ari Ramalingam, Assistant Manager (Manufacturing Engineering and New Product Development), RANE NSK Steering Systems Private Limited, Ahmedabad.
R2021	CME385 - Refrigeration and Air Conditioning	Mr.N.Bala Chandran, Proprietor, Sree Engineering. Madurai.
R20217	ME8793 - Process Planning and Cost Estimation	Mohammed Samiuddin, Deputy General Manager, TVS Training and Services Limited, Chennai.

7. Training and Placement and Laboratory Upgradation

7.1 Proposed Training Plan for Mechanical Engineering students

- Briefed the observations and outcomes of Aon Cocubes Assessment Test for 2020-2024 batch students. Identified the industry ready students based on Graduate Engineer Plant (3 Nos.), Customer Service Executive (3 Nos.), Operations executives (3 Nos.), Network Engineer (3 Nos.) and Software Tester (4 Nos.). Total Number of Students Registered for placements = 71

Proposed Training plan for Mechanical Engineering students

	First Year	Second Year	Third Year	Fourth Year
Topics Identified for Training	1. MS Office 2. Python 3. AutoCAD 4. Writing Skills 5. Quantitative Aptitude 6. Verbal & Non-Verbal Reasoning 7. Communication and Soft Skills	1. ProE/CATIA/NX CAD/Fusion 360. 2. Understanding and Reading Engineering Drawings 3. Quantitative Aptitude 4. Verbal and Non – Verbal reasoning 5. Communication & Soft Skills 6. Basics of AI/ML via Value Added Courses and NPTEL courses	1. ANSYS/ABAQUS/CFD (According to area of interest) 2. Geometric Dimensioning and Tolerancing. 3. Domain Specific Training based on Student Area of Interest. 4. Mock Interview 5. Technical Aptitude through Google Classroom 6. Basics of AI/ML via Value Added Courses and NPTEL courses	1. ANSYS/ABAQUS/CFD (According to area of interest) 2. Company Specific Training. 3. Mock Interview 4. Technical Aptitude through Google Classroom

7.2 Laboratory updation

7.2.1 Mechatronics Lab

SLNo	Name of the Equipment	Date of Inception	Cost (in Rs.)	Outcome
1	TE-A472 Aquila EV Battery Operated Vehicle for in campus application – Six seater with Lithium Battery Back	28.03.2023	4,99,999.50	To describe the electric vehicle technology
2	Y039-DMS706 YASKAWA GP12 handling Robot with YRCCON	06.12.2022	30,68,000.00	To explain the construction and working principle of real-time robots configuration and its applications
3	SIEMENS S71200 (12D1/10D0), SM1223 (8D1/8D0) Wiring	01.07.022	63,130.00	To interface the computer and the perform the TIA software/ PLC program real-time applications.

7.2.2 Thermal Lab

SLNo	Name of the Equipment	Date of Inception	Cost (in Rs.)	Remarks
1	Thermal Conductivity of Liquid Apparatus	08.09.2023	96,760.00	Upgradation of components as per Anna University Regulations
2	Critical Heat Flux Apparatus	08.09.2023	87,320.00	
3	Heat Transfer in drop wise and film wise condensation	08.09.2023	87,320.00	
4	Calibration of Thermocouple 0-80 Degree C	08.09.2023	46,020.00	

7.2.3 SOM Lab

SLNo	Name of the Equipment	Date of Inception	Cost (in Rs.)	Outcome
1	Split Die (MS), Cast Size : 300×100×15mm thick with funnel	13.03.2023	9,275.00	To perform casting process in Stir Casting machine with the split die.
2	Digital Microscope USB camera m5.0. Specification: 5.0Mega Pixel, CMOS sensor, Resolution : 2592×1944.	25.01.2023	45,000.00	To analyze the internal microstructure of the materials.
3	Metzer M Trinocular head for metallurgical microscope	25.01.2023	15,000.00	
4	Metzer M Trinocular100 X objective	25.01.2023	6,500.00	

5	Metzer M Metallurgical analysis software	Micro image	CAM	25.01.2023	76,000.00
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7.2.4 CAD/CAM Lab

SLNo	Name of the Equipment	Date of Inception	Cost (in Rs.)	Outcome
1	SAMSUNG 8GB DDR3 -RAM	23.09.2022	97,200.00	Upgradation and renewal of CAD softwares
2	Solid works platform, No of users:100, 3D CAD modelling, SIMULIA – ABAQUS, ENOVIA PLM, AR-VR Reviewer	02.03.2023	4,89,051.00	
3	EDGE CAM Software, Renewal ANSYS, PTC Creo, PTC Windchill PDM Link, No of Users: 50	27.12.2022	12,69,000.00	
4	Extension of PTC Subscription for CREO Academic Lab Pack till 27-Dec-2024, No of Users: 50	21.03.2023	3,54,000.00	

8. Continuous Improvement

Detailed to the DAC members how the new initiatives are arrived in the department and the expected outcome of these initiatives with appropriate references.

Initiatives	Drive team	Outcome	Reference
Non academic classes – Follow-up record	HOD Respective faculty I/C	Separate document or log book maintained for Non academic classes (Library/ aptitude/Placement/NM Course)	DRM/2022-23 / Even Semester / 1
Individual student assignment for R-2021	RAK & HOD	Improving student self learning & confidence	
Revision in dept. level responsibilities	HOD & All faculty members	The department level responsibilities were revised accordance with workload Devised responsibilities for dept. continuous improvements	
Co-curricular performance for attainment calculation	Respective I/C & NBA Coordinator	Suggested 1. Internship 2. Value Added Courses and 3. Industrial Visits can be included for PO attainment	DRM/2022-23/Even Semester/ 2
The new method for target calculation	Dr.SMR & HOD	Proposed course wise target calculation – For better PO attainment & transferring of Engg. attributes (NBA aspect)	
CDIO Implementation	HOD & Core team	Proposed to implement CDIO in autonomous curriculum – In progress	

Initiatives	Drive team	Outcome	Reference
Appreciation for Student/faculty	Dept. & Mr.R.Arunkumar, AP(SG)/EEE	Academic excellence posters communicated through dept. & Displayed in notice board / Honoured with rewards Faculty achievements and initiatives appreciated during DRM	DRM/2022-23 / Even Semester / 03
Domain review meeting – proposal Including students	HOD & Domain coordinators	Frequent meeting will be conducted for all domain – Course plan design/ Curricular gap identification/ Research initiatives – Frequency – one per month	DRM/2022-23 / Even Semester / 04
Categorization of students in - A, B and C	HOD / CA's / Course faculty	Customized pedagogical approach Equipping A category student as Industry ready Engineers	
Mapping of courses to be made with much more accuracy	HOD & Domain coordinators	Following "Hypothetical relevance and coverage of ABET to Bloom's 3 learning domain". DRM/2022-23 / Even Semester / 06	.
Curriculum Gap Identification – feedback from stake holders	HOD & Domain coordinators	Standard practice will be followed for all courses – Collection of course feedback from stake holders (Alumni/Industrialist/R&D Experts) & Execute the same in present semester	DRM/2023-24 / Odd Semester / 01

9. Any other matter

9.1 Transformation of Affiliated Institution to Autonomous

Detailed the progress done towards to achieve autonomous status to the DAC members

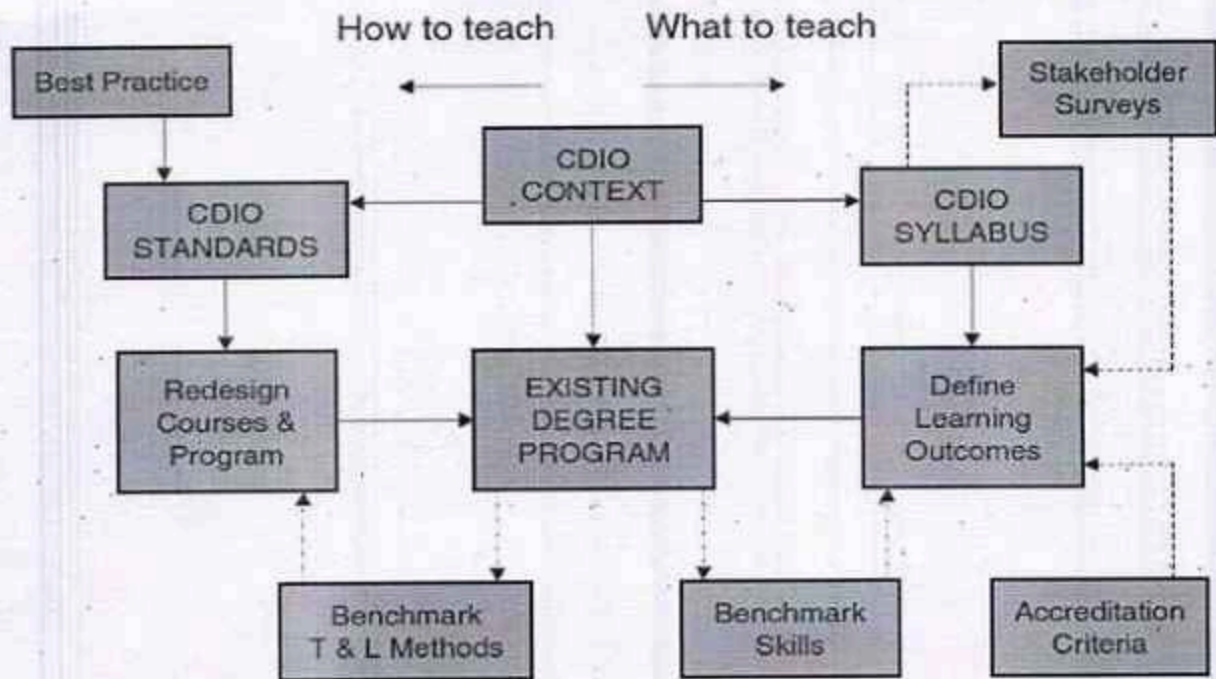
- Based on the eligibility criteria prescribed by the UGC Regulations 2023 – Notification F.No. 1-18/2021 (CPP-II) dated on 3rd April 2023 i) 10 years of existence ii) under section 2f of the UGC act (iii) Accredited by NAAC and NBA –application form is submitted in UGC web portal on 28.06.2023.
- As per the Anna University Statutes on Academic Matters 2023, the details about the submission of application form are communicated to parent university on 03.07.2023.
- For forwarding the application to UGC for Award of Fresh Autonomous Status the soft and hard copy of PROFORMA is submitted to The Director, Centre for Academic Courses, Anna University, Chennai on 19.09.2023.
- As per the bylaws, the constitution of the academic council, Board of Studies, and Finance Committees are going on.
- Curriculum Design and Developmental work towards autonomous status is going on

9.2 Implementation of CDIO framework for curriculum design and development – Initiatives:

Presented the following information related to CDIO

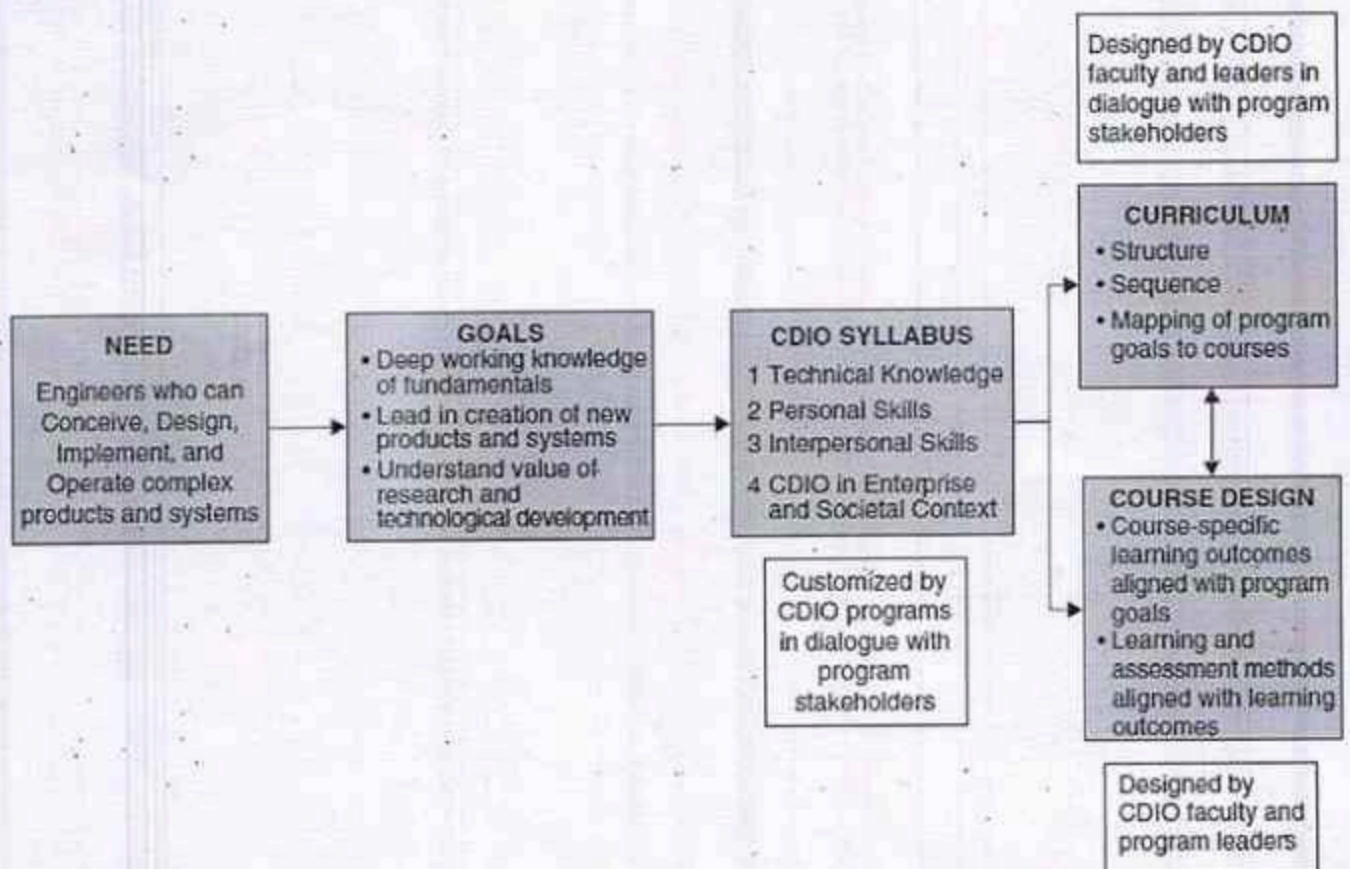
- CDIO syllabus and Standards 3.0
- Implementation of the CDIO approach

IMPLEMENTING THE CDIO APPROACH



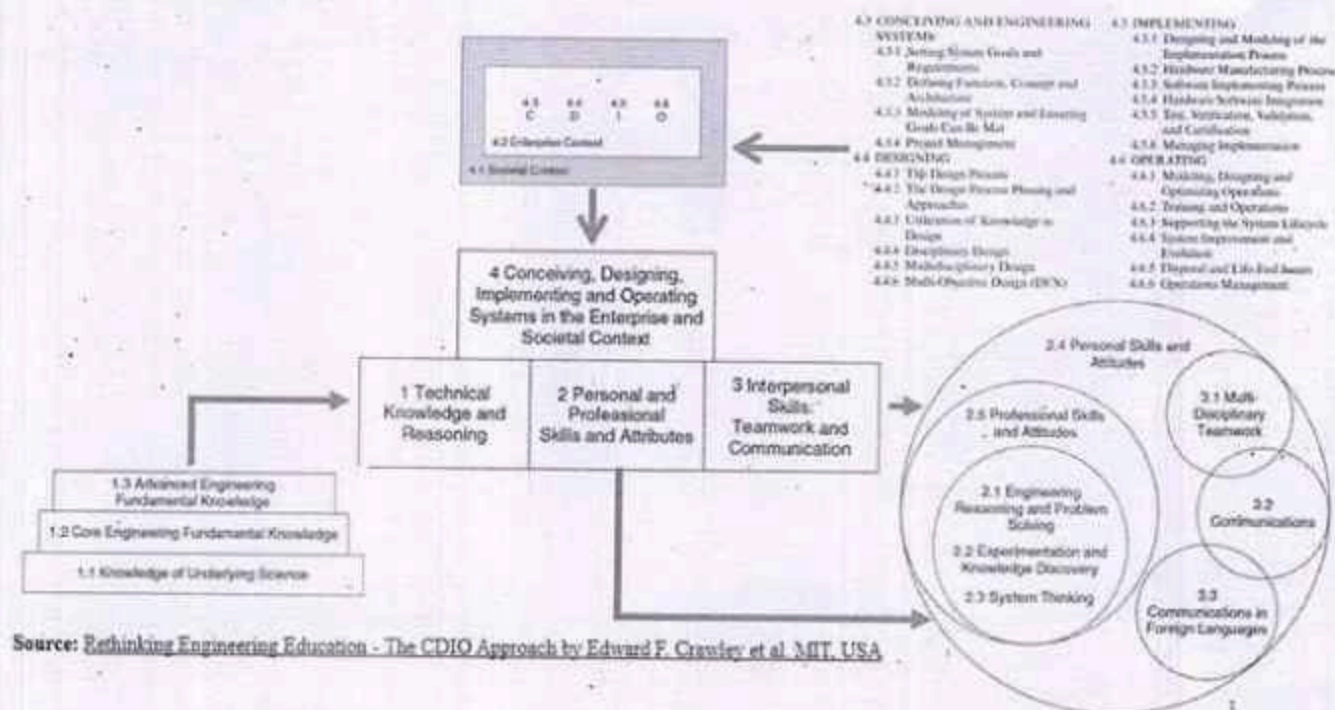
Source: Rethinking Engineering Education - The CDIO Approach by Edward F. Crawley et al.

- Development and integration of CDIO Syllabus



Source: Rethinking Engineering Education - The CDIO Approach by Edward F. Crawley et al. MIT, USA

- CDIO Syllabus @ First level



- Briefed the progress of Design Thinking course and Engineering Exploration course



9.3 Adapting functional framework for accreditation and ranking:

S.No.	Common frame work	NBA Criteria	NBA	NAAC Criteria	NAAC	NIRF Ranking	NIRF
1.	Academics	1. Vision, Mission and Program Educational Objectives (60) 2. Program Curriculum and Teaching– Learning Processes (120) 3. Course Outcomes and Program Outcomes (120)	300	1. Curricular Aspects (100) 2. Teaching, Learning and Innovation (350)	450	1. Teaching and Learning Resources (KY 1,2)	45
2.	Facilities	6. Facilities and Technical Support (80)	80	4. Infrastructure and learning resources (100)	100	1. Teaching and Learning Resources (KY 5) 4. Outreach and Inclusivity (KY 2)	45
3.	Faculty Contribution	5. Faculty Information and Contributions (200)	200	3. Research Innovations and extension (110)	110	1. Teaching and Learning Resources (KY 3) 2. Research and Professional Practices	120
4.	Student Support	4. Students' Performance (150) 8. First Year Academics (50) 9. Student Support Systems (50)	250	5. Student support and Progression (140)	140	3. Graduation Outcomes 4. Outreach and Inclusivity (KY 1,4)	150
5.	Governance	10. Governance, Institutional Support and Financial Resources (120)	120	6. Governance, Leadership and Management (100)	100	1. Teaching and Learning Resources (KY 4) 4. Outreach and Inclusivity (KY 3)	40

6.	Continuous Improvement and innovative practices	7. Continuous Improvement (50)	50	7. Institutional Values and Best Practices (100)	100	5. Perception	100
	Total	NBA	1000	NAAC	1000	NIRF	500

Suggestions given by the members

Dr.R.Jeyapaul, Prof. and Head, Department of Production Engineering, National Institute of Technology, Tiruchirappalli (Academic Member) suggestions as follows:

- Invite the industry person to deliver the content in the syllabus (preferably Unit V) based on the need.
- Offer an entrepreneurial course to all the department students by the entrepreneurs from the Department of Mechanical Engineering.
- Refer the Dr.Radhakrishnan Committee report "Transformative reforms for strengthening periodic assessment and accreditation of all higher educational institutions in India" for NBA, NAAC accreditation and NIRF ranking.
- Give more preference for student exchange programme after obtaining the autonomous status.

Dr.P.Maran, Professor, Department of Mechanical Engineering, TCE, Madurai, Academic Member. suggested the followings:

- Overview the National Education Policy and discuss in the Governing Council Meeting and take the necessary steps toward Academic Bank of Credits.
- Appreciated the mechanism adopted for the slow learner identification. Review the progress of the slow learners between the internal assessment tests.
- Motivate and encourage a greater number of students to appear for GATE exam.

Dr.K.Karthikeyan, ASCP/EEE suggested that the followings

- Whenever possible, the quantitative data can be given in the slides during the review of the actions completed.
- It is definitely commendable to put vertical in-charges for providing the honor courses. The identified faculty members can draft a specific plan and tactics for learning about those verticals.
- Internships with pay should be encouraged for students, and these internships should lead to job offers.
- The highly appreciable to see there are international participations in pursuing higher studies. The number should be increased.
- It is very encouraging that the department has a strategy in place to boost student involvement in the India Skills program. The proposed plant's execution and efficacy need to be closely watched.
- The quality of the placements should be improved. It is important to encourage and increase the number of students who choose to pursue higher education.
- Details of the numerous project proposals that the department may have submitted to the various funding organizations may have been included in the presentations.
- The department's research accomplishments deserve high praise. The department should aim for holding on-going funded projects all the time.

Mr.T.Chockalingam AP(SG)/Civil mentioned to give awareness program on govt. opportunities and IGBC professional society.

Dr.O.Senthil kumar, ASCP/Chemistry suggested that allow the students to do project work throughout the 8th semester and generate tangible outcome. The internship should not be affected the progress of the final year project.

Mr.S.P. Cheran, IV Year, Mech. appreciated the current training Plan for Mechanical Engineering students and requested to follow the same for the forthcoming batches also.

Mr.A.Arun Shenbaga Raj (Batch 2016-2020) Founder & CEO, TN76 Food Delivery Service, Tenkasi. conveyed hearty congratulations to the teams' getting grants for their projects from various Institutions and wishing all the very best in bringing their Ideas into Reality involve students in various real-time activities apart from the Syllabus.

Mr.S.Kathiravan AP/Mech. delivered the vote of thanks.



S. P. Cheran
26/06/23
NBA Coordinator / HoD/Mech.

S. Kathiravan
26/06/23
Vice Principal

M. Arun
26/06/23
Principal

Copy to:

- 1) Members of DACM
- 2) Faculty members of Mechanical Engineering through e-mail and Canvas Classroom for information and necessary action.
- 3) File